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Sleep Features on Continuous Electroencephalography Predict Rehabilitation Outcomes After Severe Traumatic Brain Injury

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Abstract

Objective: Sleep characteristics detected by electroencephalography (EEG) may be predictive of neurological recovery and rehabilitation outcomes after traumatic brain injury (TBI). We sought to determine whether sleep features were associated with greater access to rehabilitation therapies and better functional outcomes after severe TBI.

Methods: We retrospectively reviewed records of patients admitted with severe TBI who underwent 24 or more hours of continuous EEG (cEEG) monitoring within 14 days of injury for sleep elements and ictal activity. Patient outcomes included discharge disposition and modified Rankin Scale (mRS).

Results: A total of 64 patients underwent cEEG monitoring for a mean of 50.6 hours. Status epilepticus or electrographic seizures detected by cEEG were associated with poor outcomes (death or discharge to skilled nursing facility). Sleep characteristics were present in 19 (30%) and associated with better outcome (89% discharged to home/acute rehabilitation; $P = .0002$). Lack of sleep elements on cEEG correlated with a poor outcome or mRS > 4 at hospital discharge (P

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= .012). Of those patients who were transferred to skilled nursing/acute rehabilitation, sleep architecture on cEEG associated with a shorter inpatient hospital stay (20 days vs 27 days) and earlier participation in therapy (9.8 days vs 13.2 days postinjury). Multivariable analyses indicated that sleep features on cEEG predicted functional outcomes independent of admission Glasgow Coma Scale and ictal-interictal activity.

Conclusion: The presence of sleep features in the acute period after TBI indicates earlier participation in rehabilitative therapies and a better functional recovery. By contrast, status epilepticus, other ictal activity, or absent sleep architecture may portend a worse prognosis. Whether sleep elements detected by EEG predict long-term prognosis remains to be determined.

Keywords

continuous EEG; electroencephalography; modified Rankin Scale; rehabilitation; seizure; sleep; status epilepticus; traumatic brain injury

TRAUMATIC BRAIN INJURY (TBI) is a leading cause of death and disability in the first 5 decades of life, resulting in nearly 50 000 annual deaths in the United States.¹ Of those who survive moderate and severe TBI, cognitive and functional impairments occur in more than 40%, ranging from severe disorders of consciousness (such as persistent vegetative state) to more subtle cognitive disabilities.² The trajectory of recovery in the days and weeks after these brain injuries is often unclear, making it difficult to predict which patients will show meaningful neurological recovery and participate in rehabilitation. Current prognostication models include a variety of clinical and laboratory parameters, including neurological examination,³⁻⁵ radiographic data,⁶⁻⁸ and physiological metrics.^{9,10} As these models mainly focus on predicting mortality, they are unable to stratify which patients would benefit from early rehabilitation efforts. Following brain injury, clinical signs that predict long-term recovery may be obscured by the use of sedative medications, the need for operative procedures, or other interventions required in the critical care setting. Thus, the clinical examination alone in the first few days following injury is insufficient to gauge the likelihood of regaining consciousness and participating in rehabilitation. Therefore, there is a pressing need to identify objective markers that can identify patients who may benefit from early and aggressive rehabilitative treatment options.

Continuous electroencephalography (cEEG) is increasingly used in the monitoring of brain-injured patients to detect subclinical seizures, which may impair consciousness after brain injury. Continuous EEG can also be used to identify other characteristics of normal or abnormal brain function, including cerebral perfusion,¹¹ the depth of sedation/anesthesia,^{12,13} and patterns of sleep and wakefulness.¹⁴ The presence of sleep-wake cycles defines the threshold between comatose states and consciousness.¹⁵ Studies in patients undergoing rehabilitation following brain injury have suggested that the emergence of more complex sleep architecture parallels cognitive recovery¹⁶⁻¹⁹ and correlates with improvement in clinical outcome in brain-injured adults²⁰ and children.²¹ Furthermore, the presence of sleep spindles and rapid eye movement sleep in brain-injured patients in persistent vegetative or minimally conscious states predicted return to consciousness within 6 months.²² Persistent sleep disturbances are associated with a longer rehabilitation stay and poorer performance on serial neurocognitive assessments.^{17,23}

In this study, we aimed to determine (1) if elements of normal sleep architecture can be captured on cEEG in the acute phase following moderate to severe TBI and (2) if the presence or absence of sleep elements on cEEG could predict participation in rehabilitative therapies and the degree of functional recovery.

METHODS

Patient selection

This study was a retrospective chart review of patients who had suffered a TBI and were treated in the neurological or trauma intensive care unit at a single academic level 1 trauma center. Review of medical records for research purposes was approved by the Hospital of the University of Pennsylvania institutional review board. Patients were included if they were 18 years of age or older, suffered a moderate-severe TBI defined as imaging findings of TBI (including intraparenchymal contusions, subarachnoid, subdural, or epidural hemorrhage) with any Glasgow Coma Scale (GCS) score requiring ICU admission, and underwent 24 or more hours of cEEG monitoring within the first 14 days after their injury. Demographic and patient injury characteristics were obtained from the inpatient medical record. To determine the severity of neurological injury, the patient's first head computed tomographic (CT) scan was reviewed independently by 2 of the study investigators (C.W. and D.K.S.) and scored according to the Marshall and Rotterdam criteria.^{6,7}

Electroencephalographic review

Electroencephalographic (EEG) reports were reviewed for the presence of sleep elements (vertex waves, sleep spindles, and K complexes) and epileptiform activity (status epilepticus, seizures [focal or generalized], epileptiform discharges/spikes, or epileptiform activity on the ictal-interictal continuum, including periodic lateralized epileptiform discharges, stimulus-induced rhythmic periodic or ictal discharges, and frontal intermittent rhythmic delta activity). In any case where the EEG report was ambiguous, the clipped EEG record was reviewed by an independent epileptologist who was blinded to clinical information (S.S.) and specifically reviewed for sleep elements and epileptiform activity.

Outcome measurements

Primary outcome measures were (1) location of hospital disposition (home, acute rehabilitation, long-term acute care facility, or death) and (2) modified Rankin Scale (mRS) at hospital discharge. The mRS was determined retrospectively by review of physical and occupational therapy notes in the medical record. The final chart entry by the physical and/or occupational therapist, which generally corresponded to the day prior to, or the day of discharge, was used to determine the discharge mRS. Secondary outcome measurements included ICU and hospital length of stay, time to first therapy visit, and the number of visits by therapists. A favorable outcome was defined as an mRS <4 and/or discharge to home or an acute rehabilitation facility. To maintain consistency in assignment of discharge outcomes (mRS score and discharge disposition), a single researcher (D.S.) determined these outcome metrics. To minimize potential bias, EEG data were collected at a separate time point from discharge outcomes and documented on a separate spreadsheet.

Statistical analysis

Logistic regression analysis, Student 2-tailed *t* tests, and Fisher exact tests were used for statistical analyses where appropriate. Multivariable analysis using STATA software (StataCorp LP, College Station, Texas) was performed using sleep, admission GCS, and ictal-interictal activity with regard to functional outcomes (mRS). Sleep and ictal-interictal activity were analyzed as dichotomized variables (any sleep elements vs no sleep elements; any ictal-interictal activity vs no ictal-interictal activity) for the multivariable analysis.

RESULTS

Patient characteristics

Sixty-four patients met the inclusion criteria for the study (see Table 1). The majority of patients were men (53 patients) and had severe TBI based on clinical (GCS score ≤ 8) and radiographic (Marshall CT score ≥ 3 and Rotterdam score ≥ 3) criteria. The most common mechanism of injury was falls (47%), followed by motor vehicle collisions (34%) and assaults (14%). In 5% of patients, the exact mechanism of TBI could not be determined from the medical record. The majority of patients (59%) had multicompartimental injury on head CT scan and 50% required neurosurgical intervention. All patients received cEEG monitoring either for clinical concerns for seizure activity or for unexplained poor mental status. The average duration of cEEG was 50.8 hours, with a range of 24 to 168 hours.

Patient disability outcomes

Patient outcomes at hospital discharge ranged from mild disability (mRS = 2) to death (mRS = 6; Table 1). Sixteen patients (25%) died prior to hospital discharge. Of those who survived, the majority of the patients were discharged to an acute rehabilitation facility (51%), while 20% required transition to other skilled nursing care facilities. Three percent of patients were able to return home upon hospital discharge.

cEEG characteristics

Table 2 summarizes the major cEEG findings. Nine patients (14%) were found to have electrographic status epilepticus while 5 patients (8%) had electrographic seizures that did not meet criteria for status epilepticus. Another 12 patients (19%) had abnormalities on their EEG, including frontal intermittent rhythmic delta activity or other abnormalities on the ictal-interictal continuum,²⁴ including spike/wave discharges and periodic lateralized epileptiform discharges or and stimulus-induced rhythmic, periodic, or ictal discharges. Sleep features were detected in 19 patients (30%), while 45 patients (70%) did not have sleep features. In both the sleep and no-sleep groups, cEEG initiation ranged from 0 to 14 days postinjury (mean = 4.10 days $\pm$ standard error of the mean [SEM] = 1.16 for group with sleep features; mean = 3.22 days $\pm$ SEM = 0.45 for group without sleep features; *P* = .39, Student 2-tailed *t* test). Patients with and without sleep features on cEEG had similar clinical examinations at presentation as determined by the GCS (GCS score > 8 compared with GCS score ≤ 8 , *P* = .90, Student 2-tailed *t* test). Patients without sleep features tended to have more severe injuries based on radiographic features, although these differences were not statistically significant (mean Marshall score = 3.51 $\pm$ SEM = 0.24 for group without

sleep features, compared with mean Marshall score = $2.82 \pm \text{SEM} = 0.33$ for group with sleep features, $P = .11$; mean Rotterdam score = $3.33 \pm \text{SEM} = 0.15$ for group without sleep features, compared with mean Rotterdam score = $2.82 \pm \text{SEM} = 0.24$ for group with sleep features, $P = .08$, Student 2-tailed t tests).

cEEG features and patient rehabilitation outcomes

Status epilepticus was observed exclusively in patients with poor functional status, defined as an mRS ≥ 4 (see Figure 1a). Similarly, electrographic seizures were observed more frequently in patients with poor functional status, though this was not statistically significant (Figure 1a; odds ratio [OR] = 1.22; 95% CI: 0.1234-12.16; $P > .05$).

Both status epilepticus and seizures were significantly associated with death or discharge to a skilled nursing facility (Figure 2a; $P = .0395$ and $P = .0156$, respectively). Outcomes were mixed for patients who had interictal features but no electrographic seizures (see Figures 1a and 2a). Patients without seizure features tended to go home or to acute rehabilitation at hospital discharge (Figure 2a; $P = .0077$).

Compared with ictal features (including status epilepticus/seizures or abnormal discharges on the ictal-interictal continuum), the presence of sleep features was a much stronger predictor for favorable outcomes and opportunities for rehabilitation therapy. Patients in which sleep features were detected on cEEG were significantly more likely to have an mRS > 4 (Figure 1b; OR = 5.98; 95% CI: 1.49-23.93; $P = .012$) and be discharged to home or acute rehabilitation (Figure 2b; OR = 12.75; 95% CI: 2.62-62.04; $P = .0002$). Patients with sleep features were evaluated by physical and/or occupational therapists earlier in their hospital stays (9.8 days vs 13.2 days; Figure 3) and received slightly more therapy visits during their acute hospital stays (2.9 vs 3.2 visits) though these changes were not statistically significant ($P = .40$ and $P = .37$, respectively, Student 2-tailed t tests). Patients with normal sleep features tended to have shorter ICU length of stays and a strong trend toward significantly shorter length of overall hospital stay ($P = .058$; Student 2-tailed t test) (see Figure 3).

Multivariable analysis of patient outcomes

Sleep elements detected on EEG were predictive of an improved outcome (OR of mRS $\geq 4 = 0.2$; 95% CI: 0.05-0.84; $P = .03$), independent of admission GCS. When accounting for ictal-interictal activity (including status epilepticus, seizures, and interictal discharges), the presence of normal sleep architecture was independently associated with an improved outcome (OR of mRS $\geq 4 = 0.17$; 95% CI: 0.04-0.74; $P = .02$, whereas ictal-interictal activity was not. When admission GCS, sleep features, and ictal-interictal activity on EEG were included in the model, only sleep features detected by EEG were independently associated with an improved outcome (OR of mRS $\geq 4 = 0.21$; 95% CI: 0.05-0.91; $P = .04$).

DISCUSSION

This study indicates that electrophysiological patterns, including ictal, interictal, and sleep features, captured by cEEG in the acute period following TBI can provide important prognostic insights for rehabilitation and functional recovery. In this population of trauma

patients, status epilepticus and electrographic seizures were associated with a generally poor prognosis at hospital discharge (mRS ≥ 4 and/or death/skilled nursing placement at discharge). In contrast, patients who showed features of electrophysiologic sleep on cEEG had improved functional outcome at hospital discharge (mRS < 4) and were more likely to be discharged to an acute rehabilitation facility. Furthermore, multivariable analysis indicates that sleep features on cEEG are an independent predictor of favorable outcome, independent of admission GCS and the presence of ictal-interictal activity. Together, these data indicate that cEEG as a tool to detect sleep features, in addition to its standard use for seizure detection, may be useful for providing early prognostic information for clinicians, families, and patients by detecting sleep features and help stratify patients who may benefit from early and aggressive rehabilitation efforts.

Recent studies have shown that sleep disturbances are common across the spectrum of TBI, including even mild TBI.^{17,25} Sleep disturbances in patients with moderate-severe TBI are associated with longer inpatient hospital stays, a higher cost of rehabilitation, and higher rates of cognitive dysfunction and functional disability.^{26,27} These studies have largely focused on patients who are in the chronic phase of recovery after TBI. Our findings indicate that sleep patterns can be detected early after injury in some patients, and the presence of normal sleep features can provide insight into the timeline of recovery after severe TBI. Our findings are consistent with an earlier study that examined routine EEGs obtained during the “highest state of arousal” in 138 patients within 7 days after injury²⁰ and found that the presence of a sleep-like EEG record was strongly associated with a good recovery, even in patients with low GCS scores (defined as ≤ 8).

Participation in rehabilitation is imperative toward favorable long-term outcomes after TBI.²⁸ As sleep plays an important role in memory consolidation, synaptic plasticity, and neurogenesis (all processes relevant to recovery of neuronal function after injury), sufficient quality and quantity of sleep are critical during the rehabilitation process.²⁹ Furthermore, poor nighttime sleep is associated with excessive daytime sleepiness, fatigue, and anxiety, which are barriers to optimal participation in rehabilitation therapies.^{30,31}

A critical problem in the clinical care of TBI patients is the lack of evidence-based, viable therapies from large-scale clinical trials, despite what seems to be promising preclinical research.^{32,33} Part of the problem stems from the heterogeneity of the patient population and a poor understanding of the patient variables that influence long-term patient outcomes after TBI. Our study suggests that cEEG can be used in the acute setting to aid in the prediction of functional outcome after TBI. In this way, cEEG may help identify subsets of patients in future clinical trials who are most likely to benefit from early interventions that impact long-term recovery.

This study has several important limitations. First, it is a retrospective study. Because the cEEG studies were obtained as part of routine clinical care, this population likely represents a more severe end of the clinical spectrum of TBI with clinical features that supported the use of cEEG (ie, unexplained poor mental status and/or concern for subclinical seizure activity). Second, EEGs were performed at varying time points after injury, although all were in the first 14 days after injury. Third, the study did not control for the use of

medications that can influence the EEG findings, including sedative, pain, and antiepileptic medications. Generally, the use of sedating medications in these severely brain-injured patients was minimal, because of necessity for frequent neurological checks in this population, but in some cases it was required to optimize clinical care. It is possible that the use of some of these medications could have altered the EEG and obscured sleep-wake patterns. Fourth, this study assessed only patients at hospital discharge and did not include an assessment of more long-term recovery over months to years. While outcomes in our population were generally poor (53 patients had mRS ≥ 4), Cologan et al²² found that even in patients in vegetative states, EEG tracings with preserved sleep architecture were associated with improved outcomes 6 months later. Thus, these findings are consistent with our hypothesis that patients with preserved sleep architecture, but poor outcome at hospital discharge, would still be more likely to recover over the long term than those without evidence of sleep features. Future studies in which patients are followed for longer periods are warranted.

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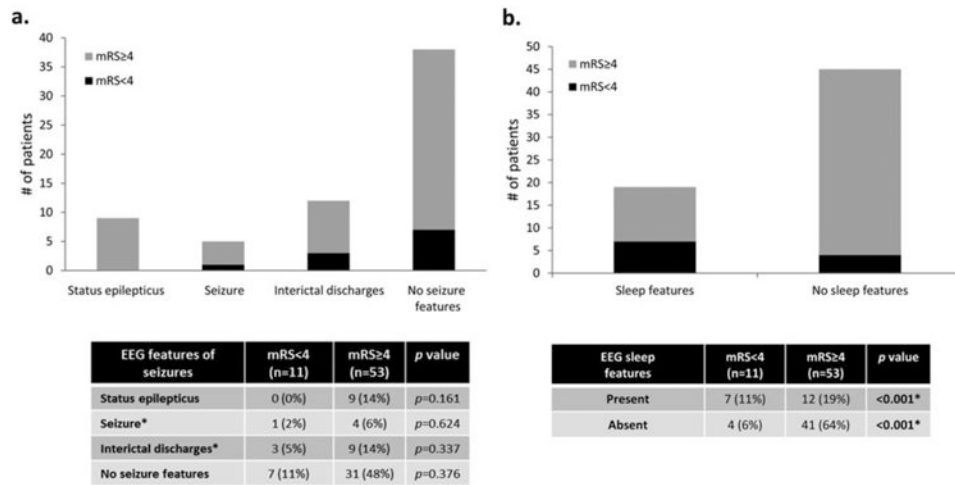


Figure 1.

Continuous electroencephalography (EEG) findings and functional outcomes. Ictal-interictal EEG patterns (panel A) and EEG sleep features (panel B) correlated with functional outcome, as determined using the mRS score. mRS <4 was considered a favorable outcome.

*A statistically significant difference between groups ($P < .05$).

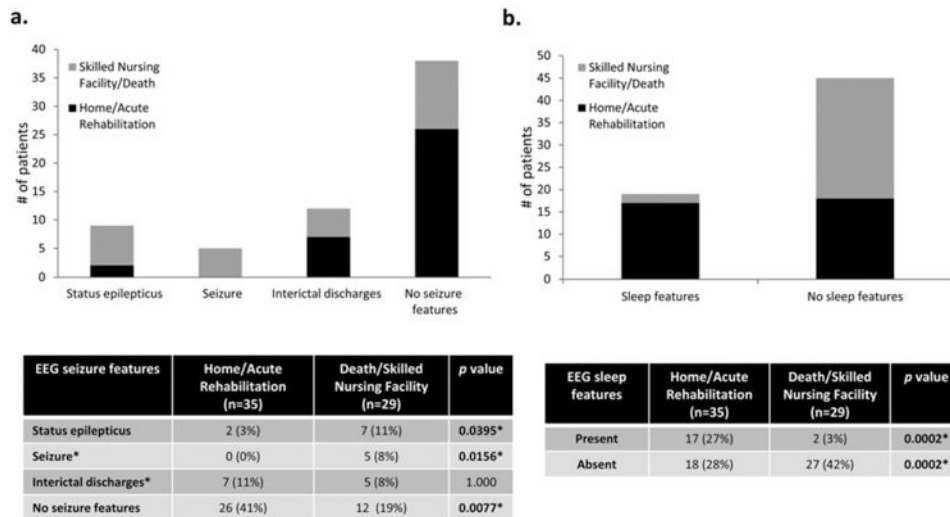


Figure 2.

Continuous electroencephalography (EEG) findings and discharge disposition. Ictal-interictal EEG patterns (panel A) and EEG sleep features (panel B) and disposition at hospital discharge. Discharges to home or acute rehabilitation were considered favorable outcomes. *A statistically significant difference between the 2 outcome groups ($P < .05$).

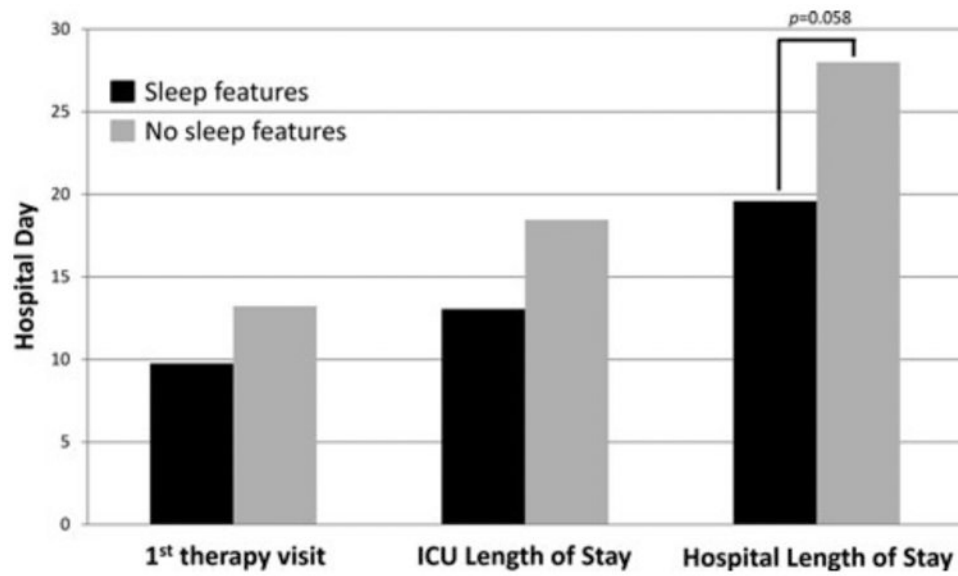


Figure 3.

Sleep features and rehabilitation time course. Number of hospital days before first inpatient therapy visit, ICU length of stay, and hospital length of stay for patients based on the presence of sleep features on continuous electroencephalography. ICU indicates intensive care unit.

TABLE 1

Patient demographics

	No. of patients (n = 64)
Gender (M:F)	53:11
Age (range)	50.3 (15-97)
Mechanism of injury	
Fall	47% (30)
MVA	34% (22)
Assault	14% (9)
Unknown	5% (3)
GCS score at hospital presentation	
≤8	59% (38)
>8	36% (23)
Unknown	5% (3)
Injury severity	
Marshall score (range)	3.4 (1-6)
Rotterdam score (range)	3.2 (1-6)
Initial head CT findings	
Multicompartment injury	59% (38)
Isolated injury	38% (24)
Unknown ^a	3% (2)
Surgical intervention	50% (32)
Hemicraniectomy	33% (21)
Hemicraniotomy	14% (9)
Burr hole	5% (2)
mRS at hospital discharge	
mRS 2	2 (3%)
mRS 3	9 (14%)
mRS 4	9 (14%)
mRS 5	28 (44%)
mRS 6	16 (25%)
Discharge disposition	
Dead	16 (25%)
Skilled care facility	13 (20%)
Acute rehabilitation facility	33 (51%)
Home	2 (3%)

Abbreviations: CT, computed tomography; GCS, Glasgow Coma Scale; MVA, motor vehicle accident; mRS, modified Rankin Scale.

^aFor some patients, initial head CT scans performed at other hospitals were not available for review.

TABLE 2

cEEG findings

cEEG findings	No. of patients (<i>n</i> = 64)
Status epilepticus	9 (14%)
Electrographic seizure	5 (8%)
Interictal abnormality ^a	12 (19%)
No epileptiform features	38 (59%)
Sleep features: present	19 (30%)
Sleep features: absent	45 (70%)

^aInterictal abnormalities include spike-wave discharges, periodic lateralized epileptiform discharges, stimulus-induced rhythmic, periodic, or ictal discharges, or frontal intermittent rhythmic delta activity.